

EWT Technology E-Commerce System

SEMESTER MARCH-AUGUST 2026

GROUP : ISP250G1

Myles Myer Marius Benedict, 2024407566,

Bryden Irywn Boilis, 2024698352,

Aiji Ellias Laison, 2024283826,

Nur Qistin Zahra binti Mohd Razali, 2024420458,

<EWT Technology E-Commerce System>	Version: <1.0>
Software Development Plan (SDP)	Date: <23/4/2026>
<Bryden Irwyn Boilis >	

Revision History

Date	Version	Description	Author
<23/4/2026>	<1.0>	<First Version of SDP is created>	<Bryden Irwyn Boilis>

<EWT Technology E-Commerce System>	Version: <1.0>
Software Development Plan (SDP)	Date: <23/4/2026>
<Bryden Irwyn Boilis >	

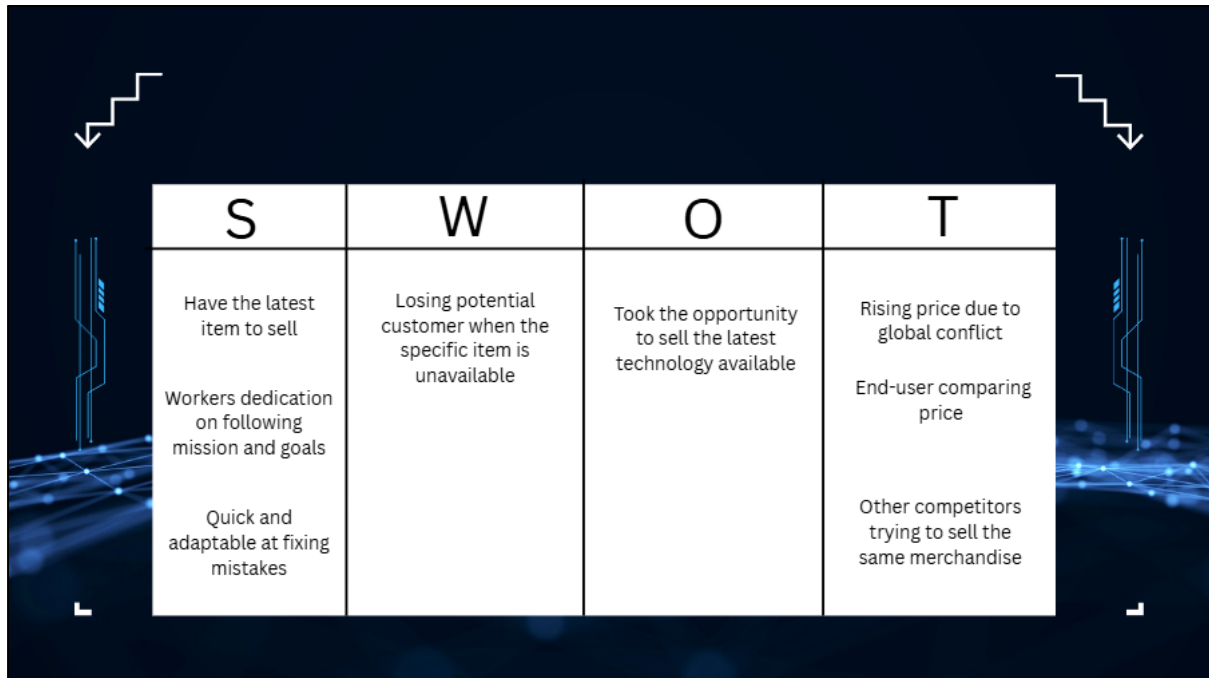
Table of Contents

1.0 INTRODUCTION	3
1.1 History of Organization	4
1.2 Organizational Chart.....	4
1.3 Current System Description	5
1.4 Current System Problem	5
2.0 PROPOSED PROJECT OVERVIEW	5
2.1 Proposed System Process	5
2.2 Proposed System Objectives	5
2.3 Proposed System Scopes.....	5
2.4 Proposed System Entity and Attributes	5
3.0 PROJECT ORGANIZATION	5
3.1 Organizational Structure	5
3.2 Roles and Responsibilities	5
4.0 MANAGEMENT PROCESS	5
4.1 Project Plan	5

<EWT Technology E-Commerce System>	Version: <1.0>
Software Development Plan (SDP)	Date: <23/4/2026>
<Bryden Irwyn Boilis >	

1.0 INTRODUCTION

1.1 History of Organization



EWT Technology is a retails company that is established in 2019. It offers consumers the latest technology available including lifestyle product. The workers in the management have 20 years of experience in retail industry. This retail store is more to ICT orientate by buying merchandise from the supplier and selling them back.

During the Covid-19 pandemic, sells hit all time low since the customer is prohibited from staying outdoors for a long period of time. The company took the initiative to do e-commerce at the marketplace such as Shoppe, Lazada, TikTok Shop and etc to reach a broader demographic consumer.

1.2 Organizational Chart

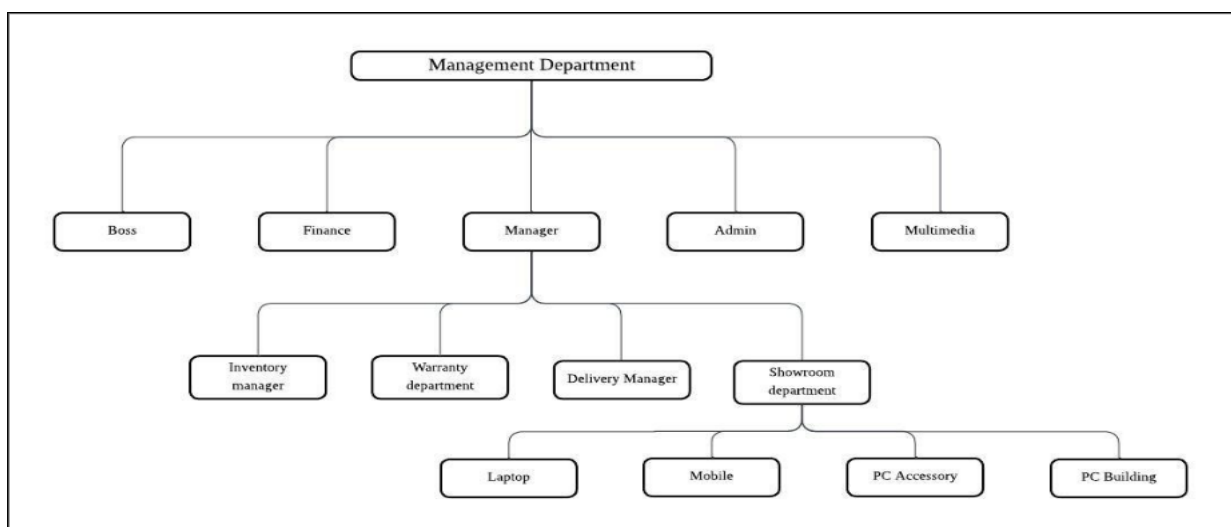


Figure 1.2.1 EWT Technology Organizational Chart

<EWT Technology E-Commerce System>	Version: <1.0>
Software Development Plan (SDP)	Date: <23/4/2026>
<Bryden Irwyn Boilis >	

The above figure shows the EWT Organizational Chart. The top layer is the management department consisting of boss, finance manager, administrator, multimedia, and manager. At second layer which is the sales department, with four distinct roles that is inventory manager, warranty manager, delivery manager. Finally, the lowest layer, showroom department, is tasked with promoting their product such as laptop, mobile, PC Accessory, PC Building.

1.3 Current System Description

Currently, EWT Technology's website is a static promotional page that lacks essential e-commerce functionalities like a shopping cart, secure checkout, and a searchable product catalogue. Because of this, customers rely on manual ordering via WhatsApp or in-store visits. Furthermore, EWT staff must process all orders, inventory, and membership rewards manually through offline data entry.

1.4 Current System Problem

The current system prevents customers from browsing specific IT components or completing purchases directly online. This disjointed and manual approach results in lost online opportunities, creates a heavy administrative workload for the staff, and significantly bottlenecks the company's potential for digital growth.

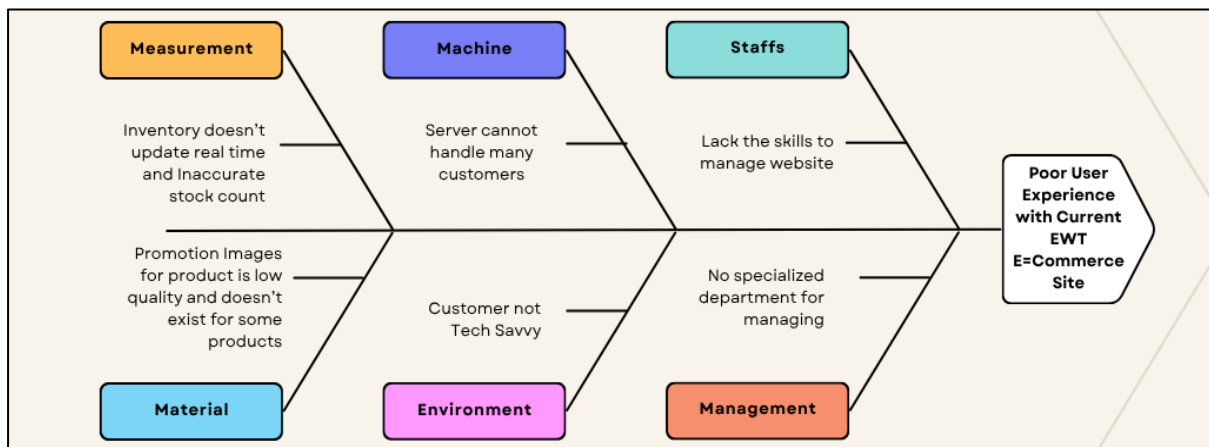


Figure 1.4.1 Ishikawa Diagram

2.0 PROPOSED PROJECT OVERVIEW

2.1 Proposed System Process

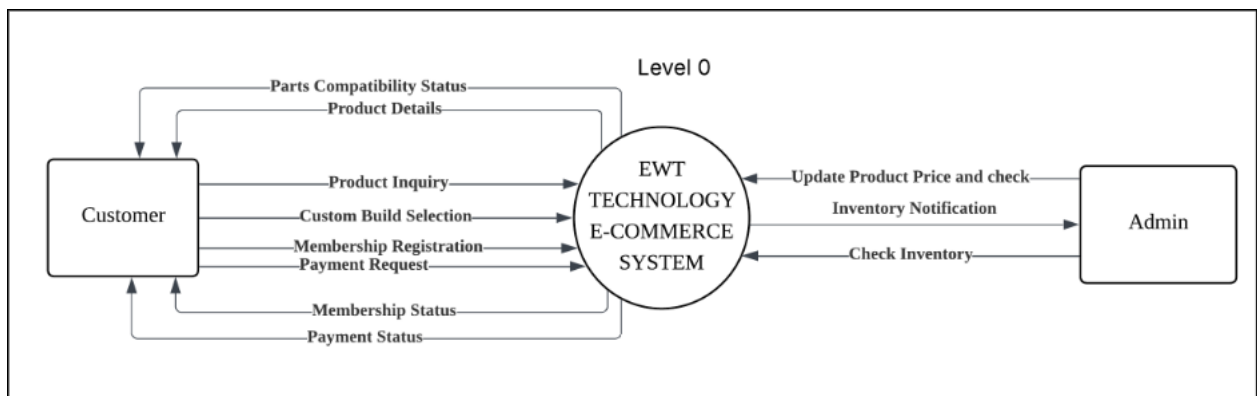


Figure 2.1.1 Context Diagram

<EWT Technology E-Commerce System>	Version: <1.0>
Software Development Plan (SDP)	Date: <23/4/2026>
<Bryden Irwyn Boilis >	

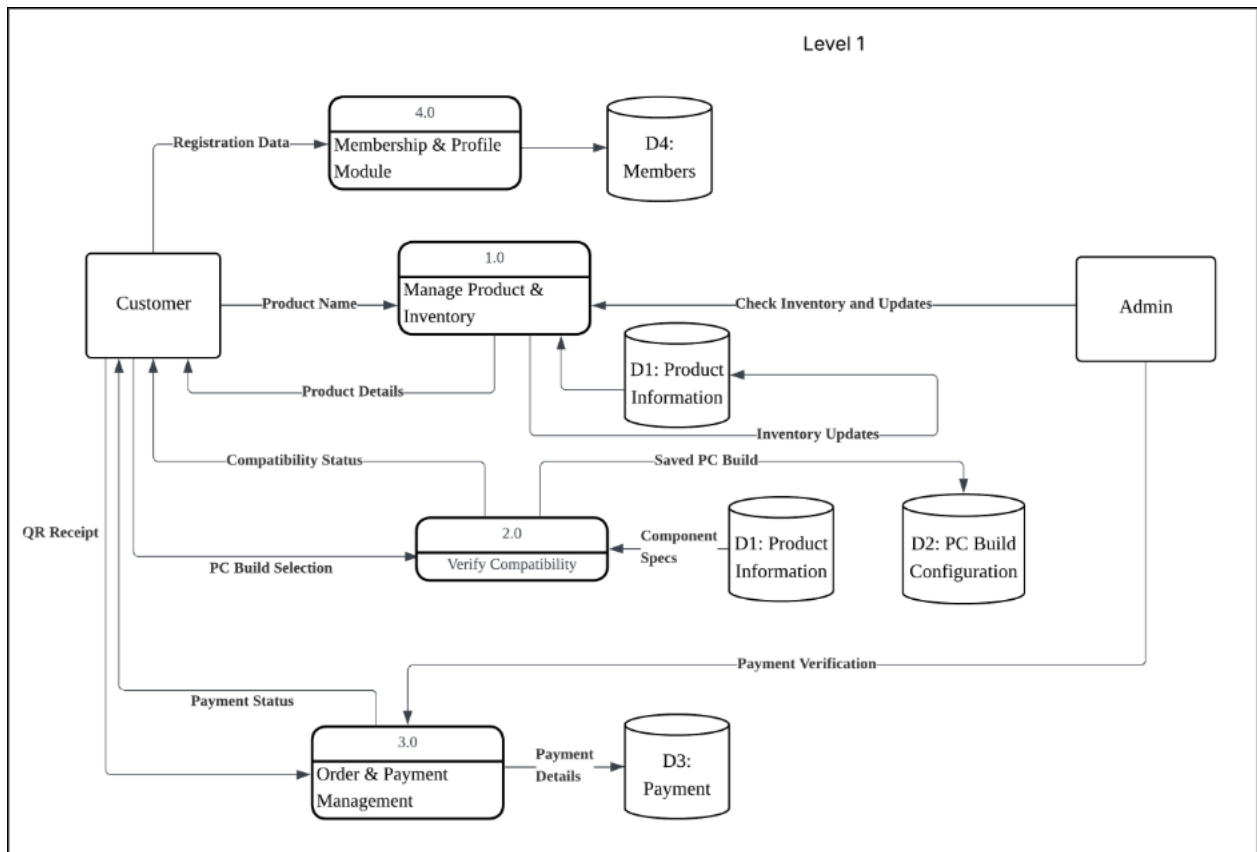


Figure 2.1.2 Data Flow Diagram Level 1

The proposed e-commerce system automates EWT Technology's workflows. Customers can browse a real-time catalogue, use a custom product configurator for IT builds, and securely check out online. The system then automatically updates inventory. Simultaneously, EWT staff will use a centralized dashboard to manage orders, track deliveries, and oversee membership rewards, eliminating the need for manual data entry.

2.2 Proposed System Objectives

- To develop a fully functional, secure e-commerce website that allows customers to independently browse products, configure IT builds, and complete purchases online
- To automate internal operations by replacing offline data entry with an integrated database for inventory management and ordering processing
- To centralize customer data, enabling automated tracking of membership tiers and reward points to improve customer retention and eliminate administrative bottlenecks

2.3 Proposed System Scopes

Target Users:

- Customers: Individuals browsing the catalogue, signing up memberships, placing orders, and tracking loyalty points

<EWT Technology E-Commerce System>	Version: <1.0>
Software Development Plan (SDP)	Date: <23/4/2026>
<Bryden Irwyn Boilis >	

- System Admin (EWT Staff): Personnel responsible for updating product listings, managing inventory thresholds, and fulfilling orders

2.4 Proposed System Entity and Attributes

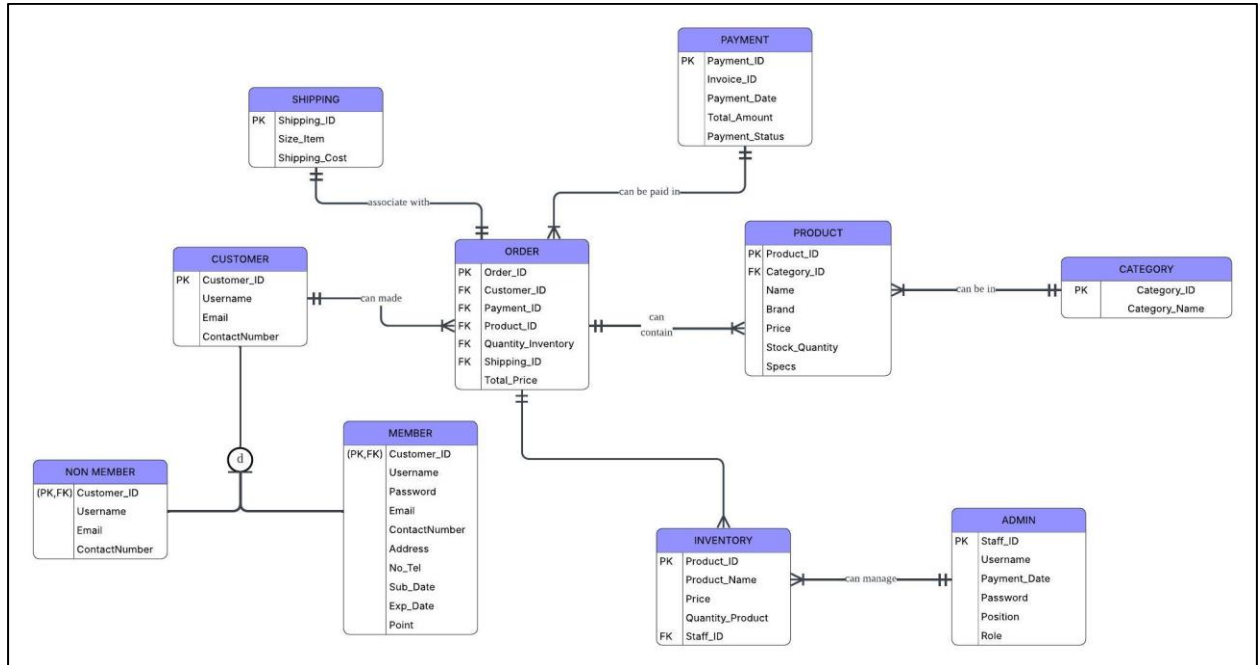


Figure 2.4.1 Entity Relational Diagram

<EWT Technology E-Commerce System>	Version: <1.0>
Software Development Plan (SDP)	Date: <23/4/2026>
<Bryden Irwyn Boilis >	

3.0 PROJECT ORGANIZATION

3.1 Organizational Structure



Figure 3.1.1 Organizational Chart

<EWT Technology E-Commerce System>	Version: <1.0>
Software Development Plan (SDP)	Date: <23/4/2026>
<Bryden Irwyn Boilis >	

3.2 Roles and Responsibilities

NAME	POSITION	JOBSCOPE
 MYLES MYER MARIUS BENEDICT	PROJECT MANAGER	• Task with project planning and scheduling • Managing resources and risk • Monitor project progress • Stakeholder communication
 NUR QISTIN ZAHRA BINTI MOHD RAZALI	SYSTEM ANALYST	• Modeling a system to visualize logic and system structure • Determine the problem requirement • Determine whether their solution is feasible
 BRYDEN IRWYN BOILIS	DATABASE ADMINISTRATOR	• Database design and implementation • Ensure security, data accuracy and validation • Optimize database to ensure fast queries
 AIJI ELLIAS LAISON	PROGRAMMER	• Testing and debugging, • Coding and documentation • Software maintenance

<EWT Technology E-Commerce System>	Version: <1.0>
Software Development Plan (SDP)	Date: <23/4/2026>
<Bryden Irwyn Boilis >	

4.0 MANAGEMENT PROCESS

4.1 Project Plan

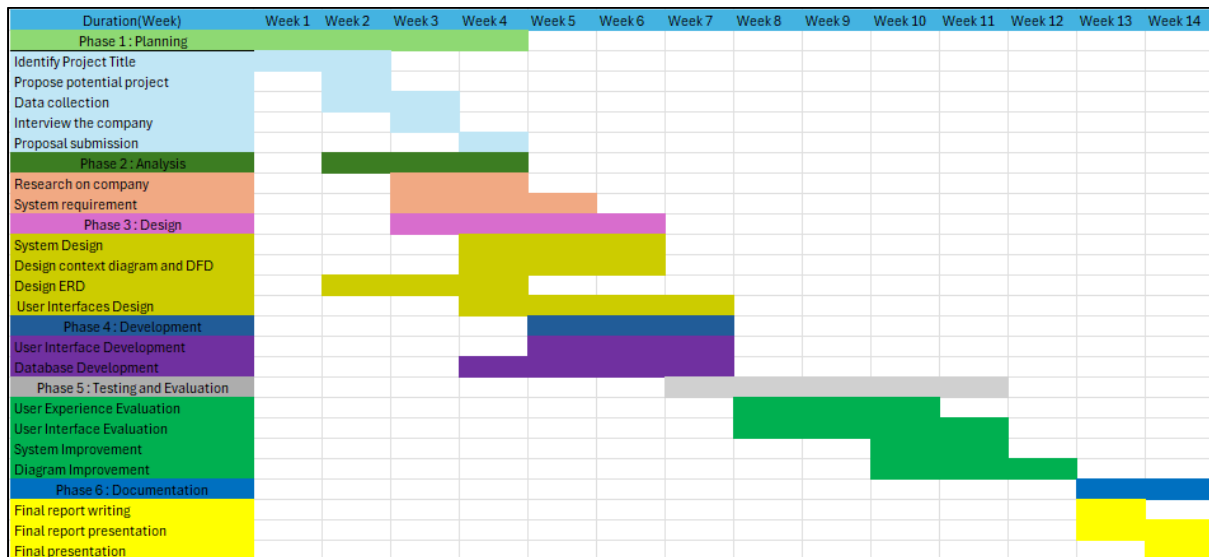


Figure 4.1.1 Gantt Chart

5.0 COMMERCIAL VALUES

1. Cost Avoidance

The system allows the organization to bypass commission rates, transaction fees, and mandatory shipping program costs simply by operating a proprietary e-commerce website. Additionally, manual labour hours can be reduced by implementing an automated order processing and inventory system. Error minimization is achieved through an integrated, real-time database that prevents costly operational mistakes, such as overselling out-of-stock merchandise or losing track of manual order records.

2. Revenue Generation

The system enables the store to transition from a static page to a functional e-commerce platform allows the business to process sales continuously, outside of standard store operating hours. These would naturally drive the upselling and cross-selling by recommending compatible, premium IT components during the build process, leading to an increased average order value. The business would also gain a wider customer base due to the online platform eliminating the geographic barriers.

3. Strategic Value

The system is centralized on analytics track purchasing behaviour and sales cycles, leading to management able to optimize inventory and marketing strategies. The gap between EWT and established industry leaders can be eliminated by shifting to a fully automated online store. Additional manpower isn't needed since the automated infrastructure allows the business to exponentially daily transaction volumes.